Quiz 1

Graded

Student

Clarisse Cheung

Total Points

1.5 / 5 pts

Question 1

Question 1 — **1** / 2 pts

1.1 a — **1** / 1 pt

- ✔ **– 0 pts** Full credit
- **– 0.5 pts** Half credit
- **– 1 pt** No credit

1 In the future, you should simplify this expression to $3/4$.

1.2 b — **0** / 1 pt

- **– 0 pts** Full credit
- **– 0.1 pts** Majority Credit
- **– 0.5 pts** Half credit
- ✔ **– 1 pt** No credit
- **– 0 pts** Incorrectly calculated P[A]
- **– 0 pts** Incorrectly calculated P[B]
- **– 0 pts** Incorrect formula used

I can't award partial credit without work shown!

Question 2

Question 2 — **0.5** / 0.5 pts

- ✔ **– 0 pts** Full credit
- **– 0.5 pts** No credit

Question 3

Question 3

0 / 2.5 pts

3.1 a

0 / 1.5 pts

– **0 pts** Full Credit

– **0.75 pts** Half Credit

✔ – **1.5 pts** No Credit

– **0 pts** did not negate the **or** correctly.

– **0 pts** did not define k.

✔ – **0 pts** did not negate the $n = 2k + 1$ statement correctly.

– **0 pts** mistakenly negated the **positive integers** statement.

✔ – **0 pts** did not negate the $\forall n$ statement correctly.

✔ – **0 pts** did not negate the $\exists k$ statement correctly.

You're missing some of the negations.

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- "For all positive integers n" is negated with the statement "There exists a positive integer n"
- The "there exists a positive integer k such that" is negated with the statement "For all positive integers k"
- The equality symbol in the statement $n = 2k + 1$ should be an $\neq$ symbol.

3.2 b

0 / 1 pt

– **0 pts** Full credit

– **0.5 pts** Half credit

✔ – **1 pt** No credit

(X or Y) is True if :

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- only X is True
- only Y is True
- both X and Y are True.

You seem to have confused OR with Exclusive OR (xor), which is False when both X and Y are True.

CMSC 28400 **Quiz 1** September 29. 2026

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Please write neatly and put your final answers in the provided boxes.

Question 1: Let $n \geq 2$ and sample $x \leftarrow \{0,1\}^{2n}$ uniformly at random. Let A be the event that x is a palindrome, and let B be the event that x starts with 00.

a) Compute $\Pr[\overline{B}]$.

$\Pr[\overline{B}] = 1 - \Pr[B] = 1 - \frac{2^{(2n-2)}}{2^{2n}}$

1

$1 - \frac{2^{(2n-2)}}{2^{2n}}$

b) Compute $\Pr[A \mid B]$.

$\Pr[A \mid B] = \frac{2^{(2n-4)}}{2^{2n}}$

$\frac{2^{(2n-4)}}{2^{2n}}$

Question 2: Let $x = 1011000$ and $y = 0110101$. Compute $x \oplus y$.

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1011000
0110101
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1101101
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1101101

Question 3 on the back.

Question 3: The following claim is false. "For all positive integers n. either n is composite or there exists a positive integer k such that $n = 2k + 1$."

a) Negate the claim.

For all positive integers n, n is both prime and there exists a positive integer k s.t. $n = 2k+1$.

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b) Disprove the claim.

Proof by contradiction. Let $k = 4$, thus $n = 2 \cdot 4 + 1 = 9$. Now consider that $9 \div 3 = 3$. Thus, $n = 9$ is both composite and equal to $n = 2k+1$ with $k = 4$.

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